



State-of-the-art review of the T/H system codes RELAP5 for HLM applications

P. Lorusso^{*}, M. Polidori, A. Del Nevo, C. Lombardo, D. Martelli, P. Meloni, M. Tarantino

ENEA, Department of Fusion and Technology for Nuclear Safety and Security, Italy

ABSTRACT

Thermal-hydraulic code RELAP5, originally developed by INEEL under NRC contract for LWRs LOCA analysis, has been extensively validated and it is actually used worldwide as a best estimate code for LWRs transient analysis. This code was chosen in the early 2000s as the reference code for the thermal-hydraulics and accident analysis of Heavy Liquid Metal Fast Reactors within the frame of Italian research programs. Several original RELAP5 routines have been updated in order to implement suitable correlations generating the reference physical and thermodynamic properties for Lead and Lead-Bismuth Eutectic fluids. Similarly, some specific heat transfer correlations for liquid metals have been added. In almost twenty years, ENEA has conducted an extensive validation program to demonstrate the capability of the code to simulate the thermal-hydraulic behaviour of heavy liquid metal systems. This program was mainly conducted on the facilities of the Brasimone center (CHEOPE, CIRCE, NACIE), but also used data from international programs (MEGAPIE International Experiment and LACANES Benchmark OECD/NEA). Furthermore, the participation of ENEA in the projects financed by the European Commission in several Euratom programs for the study and design of HLM-cooled reactors (ELSY, SEARCH, MAXSIMA, LEADER, SESAME, MYRTE) has allowed to increase the confidence on the transient and accidental analyses conducted with RELAP5 also through the comparison with the results of other codes. RELAP5 is currently a reliable tool both as a support to reactor design (safety by design) and for accidental analyses.

1. Introduction

As part of the Generation IV nuclear power plants, Lead cooled Fast Reactors (LFRs) are being developed to meet the highest requirements of safety and reliability, sustainability, economics, proliferation resistance, and physical protection. ENEA, the Italian National Agency for New Technologies, Energy, and Sustainable Economic Development is supporting the core design, safety assessment, and technological development of innovative nuclear systems cooled by Heavy Liquid Metals (HLMs) and, most recently, fully oriented on LFRs. In particular, ENEA plays a key role in the EU R&D program in both the experimental investigation on HLMs related technologies, thanks to one of the largest European fleets of experimental facilities (Lorusso et al., 2018a; Tarantino et al., 2019) at ENEA Brasimone R.C. and the development and qualification of numerical tools for specific application to HLM systems, ranging from neutronics codes, system and core thermal-hydraulic codes, computational fluid dynamics (CFD) and fuel pin performance codes, including their coupling (Tarantino et al., 2021a). Thanks to the competencies developed so far, ENEA has provided a relevant contribution in several EU projects (e.g., H2020 EC projects MAXSIMA, SESAME, MYRTE, PATRICIA, PASCAL, PIACE, GEMMA, INSPYRE,

PUMMA), as well as to the ALFRED Project in the frame of the FALCON international consortium, mainly addressing the ongoing activity in terms of core design, technology development, and auxiliary systems design.

Focusing on numerical modelling approaches and computational tools development, the Validation and Verification (V&V) process (Glaeser, 2017) is a fundamental step to demonstrate their reliability when applied to HLM technologies, for which it is necessary an extensive and well documented qualification database. To fulfill this scope, ENEA has employed R&D efforts on code modifications, model improvements, and V&V, with particular emphasis on SYStem Thermal-Hydraulic (SYSTH) codes (i.e., RELAP5), severe accident/multifluid (i.e., SIMMER) (Pesetti et al., 2015), fuel pin performance (i.e., TRANSURANUS) (Magni et al., 2020), fuel element thermal-hydraulics (i.e., ANTEO+) (Lodi et al., 2016), and computational fluid dynamic codes (CFD) (Martelli et al., 2017). Experimental data are fundamental for supporting the development and demonstrating the reliability of computer codes in simulating the behavior of a nuclear power plant, or its systems and components, during normal operation or a postulated accident scenario. For this purpose, ENEA has undertaken a series of experimental campaigns involving different experimental facilities operating

^{*} Corresponding author.

E-mail addresses: pierdomenico.lorusso@enea.it (P. Lorusso), massimiliano.polidori@enea.it (M. Polidori), alessandro.delnevo@enea.it (A. Del Nevo), calogera.lombardo@enea.it (C. Lombardo), daniele.martelli@enea.it (D. Martelli), paride.meloni@enea.it (P. Meloni), mariano.tarantino@enea.it (M. Tarantino).

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with HLMS, i.e., lead or Lead-Bismuth Eutectic (LBE), in order to create a wide database suitable for supporting the development and qualification of such computational tools.

Numerical simulations with HLMS as working fluid have been performed using RELAP5 SYS-TH code, i.e., RELAP5/Mod3.3 (RELAP5 Development Team, 1995) and RELAP5-3D© v.4.3.4 (The RELAP5-3D© Code Development Team, 2015a). In order to do this, RELAP5/Mod3.3 has been modified to simulate the LBE, Pb (Balestra et al., 2016; Oriolo, 2000; Martelli et al., 2018; Barone et al., 2019). The modifications were carried out in the framework of a larger activity aimed at setting up a code capable of dealing with HLMS technologies (i.e., implementation of liquid metals properties and related heat transfer correlations). Validation activities have been performed, starting from the LFR design and the analysis of the phenomena relevant for the simulations. Experimental data have been used to evaluate the reliability of the code results and, to some extents, also the accuracies of the predictions, as described in the present paper.

The MEGAWatt Pilot Experiment (MEGAPIE) (Zanini et al., 2011), dedicated to the design, manufacturing and testing of a liquid metal spallation target have provided relevant data from the thermal hydraulic measurements, which offered the opportunity to qualify the codes used in the design phase (i.e., RELAP5 mod3.2.2β version).

In the framework of LACANES OECD/NEA benchmark (Nitti and Meloni, 2010; Shin et al., 2014; Cho et al., 2011), RELAP5 has been used to simulate the thermal-hydraulic behaviour of the Pb-Bi loop HELIOS both in forced and in natural conditions. In the first phase of the benchmark two different forced circulation steady states have been considered in order to assess the distributed and local (based on reference models) pressure drops calculated by RELAP5. Lately, an upgraded model has been developed to characterize the natural circulation in the loop (NEA-OECD, 2018).

In the framework of the H2020 EC project SESAME, experimental data have been produced involving the NACIE-UP LBE-cooled loop-type facility (Coccoluto et al., 2011) and used to qualify RELAP5-3D© v.4.3.4 in simulating core related thermal-hydraulic phenomena (e.g., heat transfer, wall to fluid friction, transition from forced to natural circulation, and single-phase natural circulation) occurring in normal operation and accidental conditions, i.e., Protected Loss of Flow Accident (PLOFA) (Forgione et al., 2019). Furthermore, integral tests to study thermal-hydraulics and safety-related features of HLM-cooled fast reactors have been performed on the CIRCE (CIRColazione Eutettico, Eutectic Circulation) LBE-cooled pool-type facility (Lorusso et al., 2018a). The experimental outcomes have been used to assess overall performances of the RELAP5/Mod3.3 SYS-TH code at a system level (Lorusso et al., 2021a) and component level (Castelliti et al., 2019; Galleni et al., 2020), i.e., the steam generator.

2. RELAP5 qualification in MEGAPIE

MEGAPIE Target Experiment has provided data on the T/H behavior of the cooling system to support the qualification of RELAP5 mod3.2.2β version for transient and safety analysis of LBE and Lead cooling systems. The code, which was used by PSI in the design phase to simulate the behavior of the target cooling system at different operational and off-normal conditions, has been firstly assessed on the single-pin tests (Brasimone) and MEGAPIE Integral Tests (MIT) (Meloni et al., 2008). In order to correctly predict the thermal exchange in the oil side of THX (LBE-oil Target Heat Exchanger), where exchange conditions are improved by means a spiral welded on the pipe wall, the Gnielinski correlation (Gnielinski, 1976) has been implemented in RELAP5. These correlations extensively tested for curved ducts has allowed to reproduce the steady state conditions of the MEGAPIE cooling system with good agreement (see comparison reported in Table 1).

In order to verify the RELAP5 code in transient conditions, among nearly 4500 beam trips of the SINQ source recorded during the 127 days of the MEGAPIE target irradiation, a beam interruption event lasted a

Table 1

MEGAPIE Cooling System Reference Steady State.

Target Power 540.3 kW		Exp.	RELAP5 standard	RELAP Gnielinsky
THX LBE	Mflow (kg/s)*	41.23	imposed	imposed
	T1(inlet) °C	319.6	316.2	316.2
THX Oil	T2(outlet) °C	229.5	imposed	imposed
	Mflow (kg/s)	9.25	imposed	imposed
	T3(inlet) °C*	185.8	175.4	185.3
IHX Oil	T4(outlet) °C	212.9	203.5	212.6
	Mflow (kg/s)	N.A.	2.93	2.72
	T5(inlet) °C	214.4	204.1	212.9
IHX Water	T6(outlet) °C	112.0	109.4	110.7
	Mflow (kg/s)	8.04	imposed	imposed
	T7(inlet) °C	33.8	imposed	imposed
	T8(outlet) °C	49.3	50.3	50.1

sufficient time to influence the thermal-hydraulic behaviour of the cooling loop in a significant way has been recalculated. Calculated temperature trends (see Fig. 1) show the good capability of the model to predict the thermal hydraulic response of the circuit in term of the time scale. In particular, the three-way valve control, simulated in the code by means of a proportional-integral component reacts to the LBE temperature decrease with a realistic delay. An underestimation of the heat capacity of the target loop that brings to a faster cooling of the LBE in the THX component can explain slight discrepancies in the decrease of LBE and Oil temperatures.

3. RELAP5 qualification in LACANES benchmark

To provide experimental data to LACANES benchmarking, thermal-hydraulic tests were conducted by using a twelve meters high LBE integral test facility, named as HELIOS (Heavy Eutectic liquid metal Loop for Integral test of Operability and Safety of PEACER), a thermo-hydraulic scaled facility of the LBE-cooled transmutation reactor PEACER-300, constructed in Seoul National University of Republic of Korea in 2005.

The RELAP5 version used for the benchmark is the one previously qualified on the basis of the simulations performed within the experimental program carried out at ENEA Research Centre of Brasimone (Italy) on support of XADS design and MEGAPIE. As already mentioned above, the original version of the code was modified generating the physical and thermodynamics properties for Pb, Pb-Bi (soft sphere model) and for diathermic Oil and updating several original routines in order to implement new correlations for heavy liquid metals (NEA-OECD, 2015). Moreover specific heat transfer correlations were added: convective heat transfer for heavy liquid metals, evaluated according to Seban-Shimazky (pipe) or Subbotin-Ushakov (tube bundle), and for oil helical path (Gnielinsky).

The numerical model of the HELIOS loop has been developed with the same level of detail (e.g., nodalization mesh) adopted in the reactor applications, in addition, the models for the singular pressure drops computation have been drawn from an hydraulic Handbook, usually used as a reference (Idelchik, 2003). It represents a complete one-dimension description of the flow path, including also the pump bypass line utilized in natural circulation mode of the HELIOS loop.

Geometrical features of the facility and information about the conditions of the experiments are reported in (Nitti and Meloni, 2010). The calculation results have shown a quite agreement with the experimental data, confirming a general capability to reproduce the pressure drop behaviour in a liquid metal loop (see Fig. 2, left). The main drawback, because of the code mono-dimensionality nature, has resulted in the difficulty to simulate complex components, like core, valve and orifice, that are strongly dependent from tri-dimensional flow path. In these cases it has been necessary to realize a tuning of calculated pressure drop coefficient in the way to match the measured experimental values. A refine of calculation has been obtained with a calibration, on

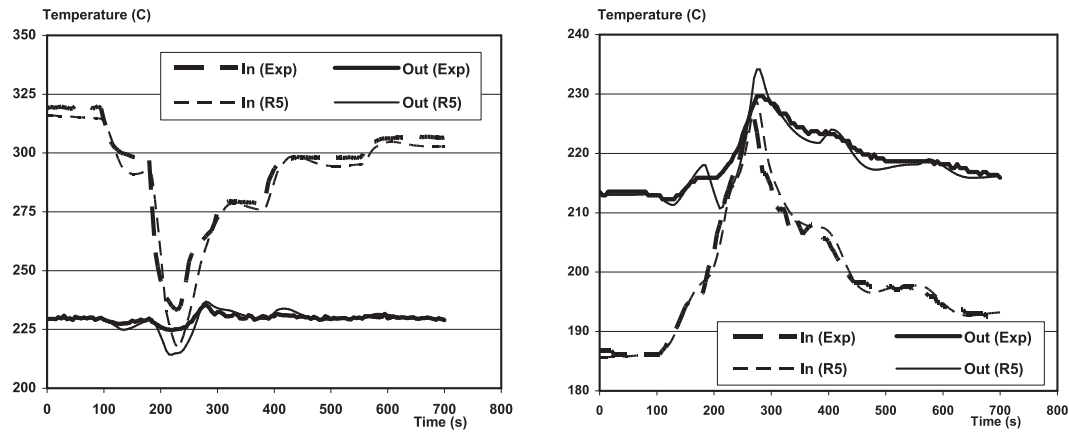


Fig. 1. THX LBE (left) and Oil (right) Temperatures.

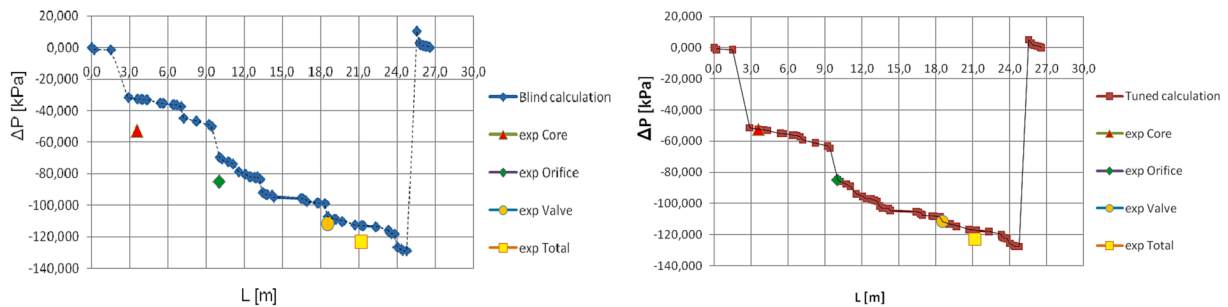


Fig. 2. Pressure drop in blind-calculation Vs experimental data (left) and in a tuned-calculation Vs experimental data for a mass flow rate of 13.57 kg/s (right).

experimental data of the concentrate pressure drop coefficients adopted for the more complex component geometries (see Fig. 2, right). Furthermore, a calculation in natural circulation has reproduced the pressure drops in the main components in quite good agreement with the pressure drop behaviour evaluated from measured experimental data (see Fig. 3).

4. RELAP5 qualification in NACIE-UP loop-type facility

Within the framework of HORIZON 2020 Safety Assessment of Metal cooled reactor (SESAME) project, a test campaign was performed with the NACIE-UP (NAtural Circulation Experiment- UPgraded) facility, hosted at the ENEA Brasimone Research Centre, in order to achieve experimental feedback on HLM thermal-hydraulics and to support the

qualification and validation process of numerical tools for HLM simulation. NACIE-UP (Di Piazza et al., 2016) is a LBE loop facility to qualify and characterize components, systems and procedures relevant for HLM nuclear technologies. It is a rectangular loop (7.7 m height) consisting of two vertical pipes (O.D. 2.5", S40), namely the downcomer and the riser, connected with two horizontal pipes (O.D. 2.5", S40). In the lower part of the riser a prototypical wire-spaced fuel pin bundle simulator (FPS) consisting of 19 wire-spaced electrical pins, arranged in a triangular lattice is installed, whereas a heat exchanger (HX) is placed in the upper part of the downcomer. The difference in height, H , between the centre of the FPS and the centre of the HX is about 5.5 m ensuring the driving force to sustain natural circulation inside the loop. An argon gas injection device is placed inside the riser to promote the circulation inside the loop. An expansion vessel is installed, coaxially with the riser (on the top

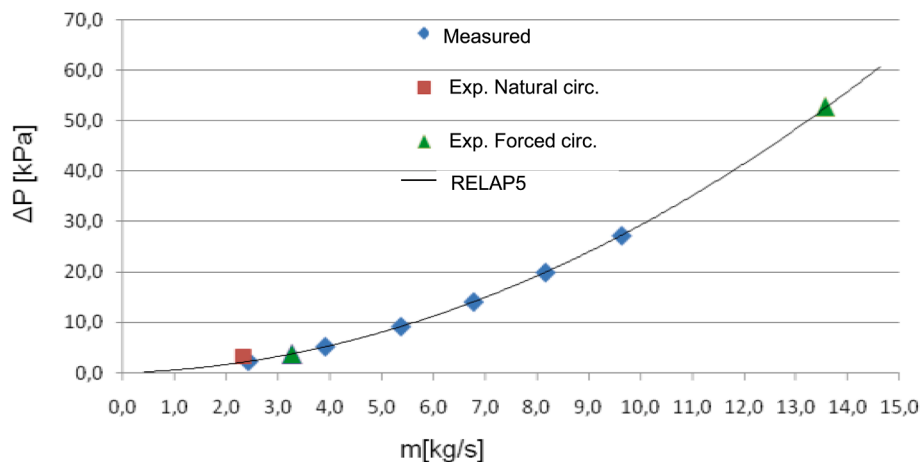


Fig. 3. Core pressure drop Vs mass flow.

part), enabling the thermal expansion of the LBE during operational transient and allowing the separation of the argon from the LBE. The experimental campaign consisted in three transients (operative and accidental) relevant for HLM nuclear systems:

1. Gas flow transition (Test-1). It consists in a reduction of the injected argon flow from 20 to 10 NL/min, maintaining the FPS power at 50 kW. The HX water inlet temperature is set to 170 °C and the pressure at 16 bar. The water mass flow rate was maintained constant at 10 m³/h during the whole test.
2. Power transition (Test-2). It consists in a FPS power reduction from 100 to 50 kW (decreasing rate 1 kW/s) maintaining the injected argon flow to 18 NL/min. The HX water inlet temperature is set to 170 °C and the pressure at 16 bar, while the water mass flow rate was maintained constant at 6.6 m³/h.
3. Protected Loss of Flow Accident, PLOFA (Test-3). This kind of transition reproduces a protected loss of flow caused by the removal of the gas lift enhancing the loop circulation and the establishment of natural circulation. In particular, it consists in a decrease of the FPS power from 100 to 10 kW (decreasing rate of 10 kW/s) and the complete deactivation of the injected argon flow from 20 to 0 NL/min. In this case, the water inlet conditions are identical to those of Test-1.

With the aim to support the qualification process of RELAP5 for HLM reactors, a numerical model of the loop-type facility NACIE-UP has been developed by means of RELAP5-3D© ver4.3.4 (The RELAP5-3D Code Development Team, 2015b, The RELAP5-3D Code Development Team, 2015c) and used in a post-test analysis for the simulation of three experimental tests described above, comparing the numerical results obtained with the experimental data available from the experimental campaign (Lorusso et al., 2018b). RELAP5-3D hydrodynamic model matches in a reliable way the experimental behaviour, with a good agreement between the LBE mass flow rate computed by the code and the one measured by the thermal flow meter. The transient trends for the three tests are well defined and the main phenomena have been well reproduced, even if in sudden transients (see Fig. 4).

The temperature trends of LBE obtained by the code are very similar to the experimental ones, both for steady state and transient conditions. A discrepancy between the computed and experimental temperatures ranges has been noticed, for both FPS and HX, with the numerical values which underestimate the experimental data. The sensitivity analysis carried out on the stainless steel powder thermal conductivity led to a good reproduction of the experimental temperature trends, reducing in relevant way the discrepancy between the computed and experimental temperatures ranges (Forgione et al., 2019).

5. RELAP5 qualification in CIRCE-hero pool-type facility

The LBE pool CIRCE (CIRcolazione Eutettica) is an integral effect facility set at ENEA CR Brasimone. Its main features, along with the implemented HERO test section geometry and instrumentation installed

were detailed in Lorusso et al., 2019. The facility consists of a cylindrical main vessel (1.2 m in diameter and 8.5 m in height), with 9.0 m³ of LBE, hosting ad-hoc test sections. A Fuel Pin Simulator (FPS) with 37 electrically heated rods in a hexagonal array with a total power up to 1 MW is placed at the lowest position, while the heat exchanger is placed at the top (Lorusso et al., 2019). The LBE circulation due to buoyancy (given the height difference between heat source and sink) can be further enhanced by gas (argon) injection in the riser section.

In the framework of the HORIZON2020 SESAME European project, the HERO test section (Lorusso et al., 2018c) has been designed, including a Steam Generator Bayonet Tube (SGBT) as main cooler, representing the ALFRED steam generator (mock up 1:1 in length). The component consists of a tube bundle of seven double-wall bayonet tubes arranged with a triangular pitch in a hexagonal shell. When the facility is in operation, the LBE contained inside the pool is heated in the FPS and it reaches the separator on the top of the test section passing through the riser. Then, LBE flows down crossing the shell side of the tube bundle for six meters, leaving the component from the bottom. The SGBT is fed with demineralized water up to 180 bar and 335 °C, reproducing the conditions of the ALFRED SG.

In this configuration, a high-pressure experimental campaign has been performed. The test matrix (Lorusso et al., 2019) is composed of three experimental transient tests aiming at reproducing Protected Loss Of Flow Accident scenarios (PLOFAs). The tests allow to evaluate the thermal-hydraulic performance of a HLM pool-type facility when an accidental scenario occurs, and to achieve experimental data relevant for the ALFRED SG and codes validation. In the first test, power decreased following the decay heat curve, gas lift was disabled in 10 s and feedwater to 30 % mass flow rate (simulating DHR system) in 2 s. The second test differs from the first one only for the feedwater reduction to 0 % in about 2 s (without DHR). The third test simulated the power decay curve, DHR (feedwater to 30 % in 2 s) and reactor pump flywheel by a gas lift reduction based on a defined table.

The nodalization of the CIRCE-HERO facility (Lorusso et al., 2021a) has been set-up by up-grading the former CIRCE-ICE nodalization (Tarantino et al., 2012), thus keeping the same basic modelling approach, using RELAP5/Mod3.3, modified to simulate the LBE (Balestra et al., 2016; Oriolo, 2000; Martelli et al., 2018; Barone et al., 2019), i.e., with the implementation of liquid metals properties and related heat transfer correlations). The model can be divided into three main parts: the pool zone, the main circulation zone (including the FPS, the riser and the LBE side of HERO test section), and the water side of the HERO test section. The elevations of the different parts of the facility are maintained in the nodalization. A sliced approach is applied at all zones, LBE and water systems. The node to node ratio is kept uniform with a maximum ratio of 1.2 between adjacent sub-volumes. The heat transfer correlations applied in the LBE zone are Ushakov in the bundle and Seban-Shimazaki, elsewhere. Further details of the components modelling are reported in Lorusso et al., 2021a.

The imposed boundary conditions have been implemented in the model and the steady state results achieved are compared with experimental data in Table 2. In particular, the transient test #3 (named SE-

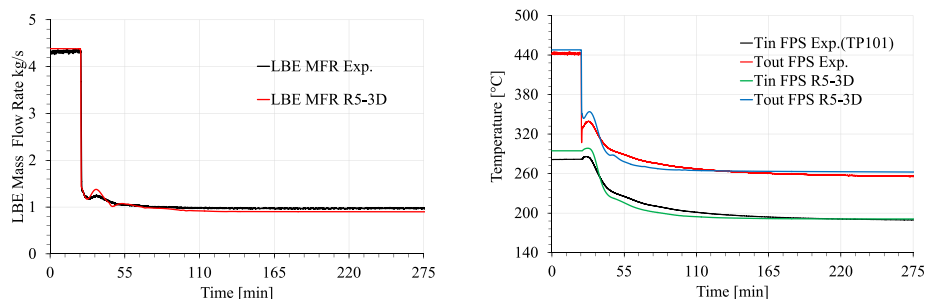


Fig. 4. Test 3 (PLOFA) LBE mass flow rate (left) and FPS temperatures (right), experimental Vs. RELAP5-3D©.

Table 2
RELAP5/Mod3.3: steady state results.

QUANTITY	UNIT	Exp	Calc
Core thermal power	kW	356	356.0 (0.0 %)
HERO power exchanged	kW	–	351.6 (–)
Pool vessel heat losses	kW	–	37.3 (–)
Feeding conduit and riser – pool heat exchange	kW	–	18.7 (–)
Pool cover gas	MPa (g)	0.014	0.02 (42.9 %)
Secondary system steam line pressure	MPa (g)	17.1	17.2 (0.6 %)
LBE FPS inlet	°C	419.6/419.4/431.7	419.6 (0.1 °C)
LBE FPS outlet	°C	495.5/497.8/495.2	493.2 (–3.0 °C)
LBE HERO inlet	°C	470.3/485.3/485.3	489.4 (4.1 °C)
LBE HERO outlet	°C	395.7/413.4/400.9	417.4 (14.0 °C)
Secondary side DWBT SG inlet (manifold zone)	°C	336.7	336.0 (–0.7 °C)
Secondary side DWBT SG outlet (steam line zone)	°C	356.7/357.8/354.7	365.3 (8.9 °C)
Ar injection	nl/s	2.75	2.72 (–1.1 %)
Primary system (venturi)	kg/s	33.3	33.6 (0.9 %)
FW total	kg/s	–	0.308 (–)

TEST3) (Lorusso et al., 2019) has been assumed as reference test for a numerical exercise.

At time of the Start of Transient (SoT), the LBE coolant temperature at the FPS inlet is equal to the experimental measurements. The LBE temperature at the FPS outlet is 3 °C lower than the experimental measures, which is considered acceptable, considering that it corresponds to a difference of about 3.9 % (TCs precision $\sim \pm 1.5$ °C). This implies that the difference can be justifiable if the uncertainty of the LBE fluid proprieties (NEA-OECD, 2015) and the accuracy in measuring the LBE mass flow rate are considered. Larger difference is observed considering the temperature at HERO test section inlet (i.e. + 4.1 °C in the calculation results). Indeed, the temperature of the coolant exiting from the FPS and flowing along the fitting volume and the riser drops of about 11 °C during the experiment, while it decreases of only 4 °C in the RELAP5/Mod3.3 simulation due to the effect of the energy removed by the air cooling system operated in the dead volume (i.e., the component holding the FPS electrical connections), as well as the effect of the heat exchange between the fitting volume and the riser and the pool zone. Comparing the LBE temperatures at HERO outlet the difference between the experimental datum and the calculation result is more relevant (i.e. about –14 °C). However, the experimental value considered for the comparison is the average value between the 3 thermocouples having a relevant scattering between the maximum and the minimum values. The overall calculated values are considered acceptable to proceed with the transient simulation.

Transient is obtained by implementing in the model the sequence of events performed during the experiment. The results are qualitatively reported in the figures below. The mass flow rate of the code simulation predicts reasonably the experimental time trends (see Fig. 5), and the differences are easily connected with the real layout and behavior of the gas injection line of the facility (i.e. length > 15 m after the mass flow measurement, having the last part of the gas pipe descending inside the pool) reproduced in a simplified way in the code nodalization. In long term, the natural circulation is stabilized.

The FPS inlet and HERO inlet/outlet LBE coolant temperatures are equivalent to the experimental data time trends. On the opposite, the coolant temperature downstream the heated part (Fig. 6) drops smoother than in the simulation. This effect can be explained with a more effective thermal coupling between the coolant flowing in the rising channel and the pool temperature, excluding the possibility that the nodalization under-estimates the thermal inertia of this zone. Looking the coolant temperatures in the HERO test section, these are apparently well predicted by the code simulation, considering that differences observed at SoT. and highlighted above are reduced during this phase. Nevertheless, a slower temperatures increase is observed at the beginning of the transient. On the opposite, the temperature increase of the water steam in HERO secondary side is over-predicted by the code (see Fig. 7). Finally, the temperature profile in the pool is reported in Fig. 8. Among the TCs supporting rods installed in the pool, the rod “line A” has been reported in this paper, since it is the one with the higher number of TCs installed. The temperature vs height trend is reported at three different times, 0 s, 800 s and 1400 s. The analysis of experimental data (Lorusso et al., 2019) shows that the pool temperature results approximately uniform in each horizontal section. The profile is reasonably predicted by the code simulation, notwithstanding the transition between the low and high temperature transition zones is placed about 30 cm above.

6. Conclusions

The goal of the paper is to present the state of the art of the RELAP5 system code qualification at ENEA throughout the description of the results obtained in some of the main projects financed by the European Commission in several Euratom programs for the study and design of HLM-cooled reactors. The RELAP5/Mod3.3 code underwent to modifications involving the implementation of suitable correlations generating the reference physical and thermodynamic properties for Lead and LBE fluids. Similarly, some specific heat transfer correlations for liquid metals have been added. To assess and to qualify the capabilities of the STH codes in simulating the thermal-hydraulic behaviour of Gen-IV related HLM-cooled systems, several experimental data coming from different HLM-cooled facilities have been used. The comparison between the experimental data and numerical simulations showed a good capability of the code in simulating the main phenomena occurring in both loop-type facilities (e.g., NACIE-UP) and pool-type facilities (e.g.,

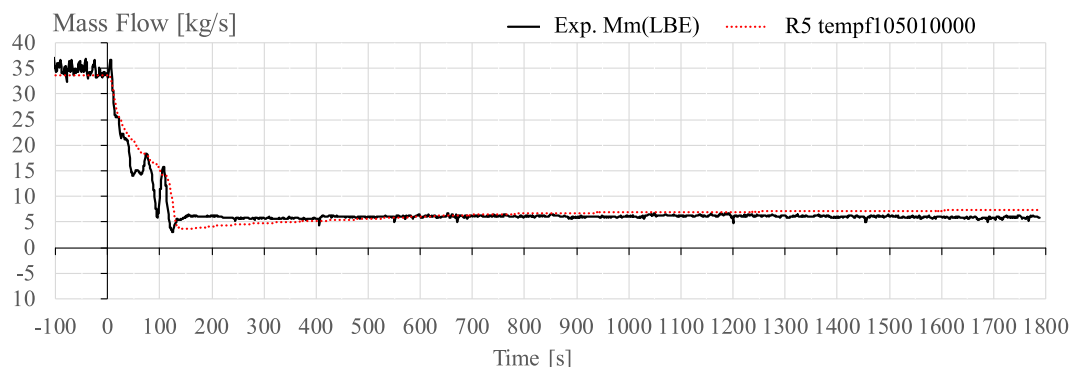


Fig. 5. LBE coolant mass flow rate – experimental data vs. calculated time trends.

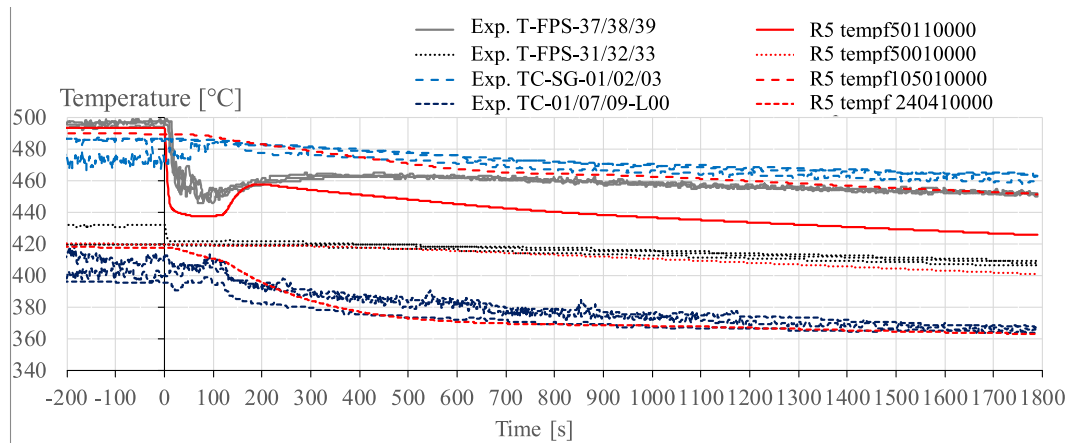


Fig. 6. LBE coolant temperatures @ FPS inlet and outlet, @ HERO inlet and outlet – experimental data vs. calculated time trends.

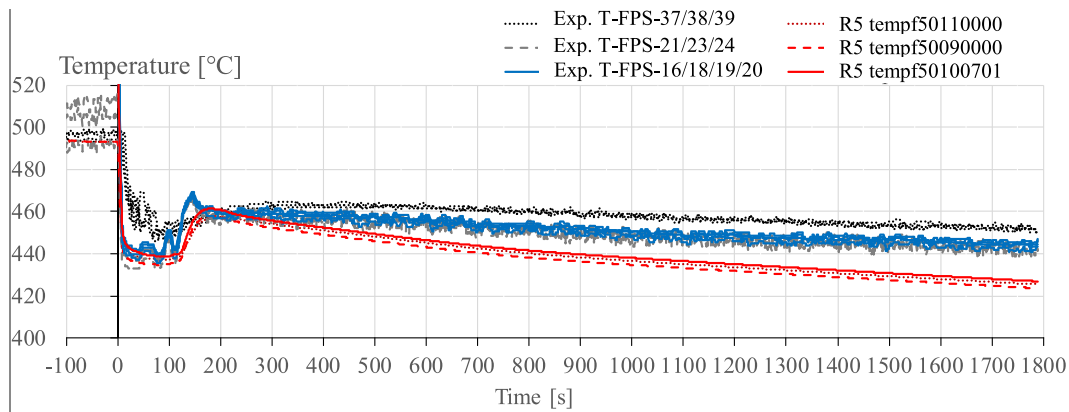


Fig. 7. Steam coolant temperature at BT outlet – experimental data vs. calculated time trends.

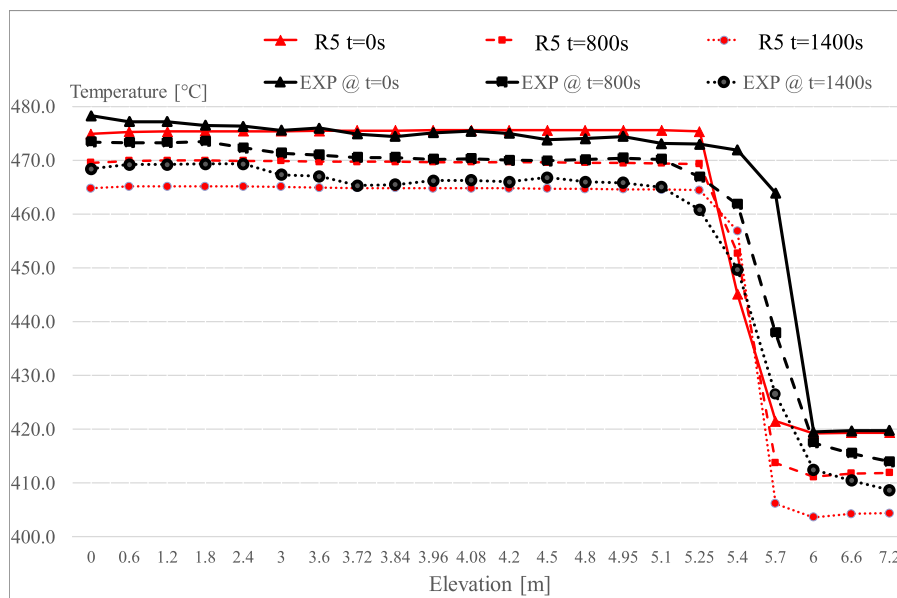


Fig. 8. LBE coolant temperatures @ pool Line A – experimental data vs. calculated time trends.

CIRCE), reproducing with good accuracy the time trends of the main operating parameters (e.g., mass flow rate, temperature, pressure) during the steady state conditions (forced and natural circulation

regimes) and during postulated accidental transients (i.e., PLOFA).

Currently, further experimental campaigns are foreseen at ENEA Brasimone Research Centre, where a new test section for the CIRCE

facility is under commissioning (Lorusso et al., 2021b; Tarantino et al., 2021b). The new experimental device will be used to produce relevant experimental data suitable to support the code verification and validation process in simulating pool thermal-hydraulics phenomena, thermal mixing, multidimensional coolant temperatures, flow distributions and heat transfer in prototypical components for HLM pool-type reactors.

CRediT authorship contribution statement

P. Lorusso: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **M. Polidori:** Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – review & editing, Supervision. **A. Del Nevo:** Methodology, Investigation, Formal analysis, Data curation, Funding acquisition, Project administration, Writing – review & editing, Supervision. **C. Lombardo:** Methodology, Investigation, Writing – review & editing. **D. Martelli:** Methodology, Investigation, Writing – review & editing. **P. Meloni:** Methodology, Investigation, Visualization, Writing – review & editing, Supervision. **M. Tarantino:** Writing – review & editing, Funding acquisition, Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Balestra, P., Giannetti, F., Caruso, G., Alfonsi, A., 2016. New RELAP5-3D lead and LBE thermophysical properties implementation for safety analysis of Gen IV reactors. *Sci. Technol. Nucl. Install.* 2016 (2016), 1687946.
- Barone, G., Martelli, D., Forgione, N., 2019. Implementation of Lead-Lithium as working fluid in RELAP5/Mod3.3. *Fusion Eng. Des.* 146, 1308–1312.
- Castelliti, D., Hamidouche, T., Lorusso, P., Tarantino, M., 2019. H2020 MYRTE CIRCE-HERO experimental campaign post-test activity and code validation, 18th International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-18), Portland, Oregon, August 18–23, 2019, 2-s2.0-85073733801.
- Cho, J.H., Batta, A., Casamassima, V., Cheng, X., Choi, Y.J., Hwang, I.S., Lim, J., Meloni, P., Nitti, F.S., Dedul, V., Kuznetsov, V., Komlev, O., Jaeger, W., Sedov, A., Kim, J.H., Puspitarini, D., 2011. Benchmarking of thermal hydraulic loop models for Lead-Alloy Cooled Advanced Nuclear Energy System (LACANES), phase-I: Isothermal steady state forced convection. *J. Nucl. Mater.* 415 (3), 404–414.
- Coccoluto, G., Gaggini, P., Labanti, V., Tarantino, M., Ambrosini, W., Forgione, N., Napoli, A., Oriolo, F., 2011. Heavy liquid metal natural circulation in a one-dimensional loop. *Nucl. Eng. Des.* 2011 (241), 1301–1309.
- Di Piazza, I., Angelucci, M., Marinari, R., Tarantino, M., Forgione, N., 2016. Heat transfer on HLM cooled wire-spaced fuel pin bundle simulator in the NACIE-UP facility. *Nucl. Eng. Des.* 300, 256–267. <https://doi.org/10.1016/j.nucengdes.2016.02.008>.
- Forgione, N., Martelli, D., Barone, G., Giannetti, F., Lorusso, P., Hollands, T., Papukhiev, A., Polidori, M., Cervone, A., Di Piazza, I., 2019. Post-test simulations for the NACIE-UP benchmark by STH codes. *Nucl. Eng. Des.* 2019 (353), 110279 <https://doi.org/10.1016/j.nucengdes.2019.110279>.
- Galleni, F., Barone, G., Martelli, D., Pucciarelli, A., Lorusso, P., Tarantino, M., Forgione, N., 2020. Simulation of operational conditions of HX-HERO in the CIRCE facility with CFD/STH coupled codes. *Nucl. Eng. Des.* 361 (2020), 110552 <https://doi.org/10.1016/j.nucengdes.2020.110552>.
- Glaeser, H., 2017. 13-Verification and validation of system thermal-hydraulic computer codes, scaling and uncertainty evaluation of calculated code results, in F. D'Auria. *In Thermal-Hydraulics of Water Cooled Nuclear Reactors*; Woodhead Publishing: Cambridge, UK, 2017; pp. 831–903.
- Gnielinski, V., 1976. New equations for heat and mass transfer in turbulent pipe and channel flow. *Int. Chem. Eng.* 16 (1976), 359–368.
- Idelchik, I.E., 2003. *Handbook of Hydraulic Resistance*, 3rd ed. Published by Jaico Publishing House.
- Lodi, F., Grasso, G., Mattioli, D., Sumini, M., 2016. ANTEO+: A subchannel code for thermal-hydraulic analysis of liquid metal cooled systems. *Nucl. Eng. Des.* 301, 128–152.
- Lorusso, P., Del Nevo, A., Narcisi, V., Giannetti, F., Caruso, G., Zwijsen, K., Breijder, P.A., Hamidouche, T., Castelliti, D., Rozzia, D., Tarantino, M., 2021a. Total Loss of Flow Benchmark in CIRCE-HERO integral test facility. *Nucl. Eng. Des.* 376, 111086 <https://doi.org/10.1016/j.nucengdes.2021.111086>.
- Lorusso, P., Bassini, S., Del Nevo, A., Di Piazza, I., Giannetti, F., Tarantino, M., Utili, M., 2018a. GEN-IV LFR development: status & perspectives. *Progress in Nuclear Energy* 105, pp. 318–331. (2018). <https://doi.org/10.1016/j.pnucene.2018.02.005>.
- Lorusso, P., Giannetti, F., Caruso, G., 2018b. NACIE-UP RELAP5 / RELAP5 3D simulations unioma1, WP 5.4: NACIE-UP benchmark activity, March 2018.
- Lorusso, P., Pesetti, A., Tarantino, M., 2018c. ALFRED steam generator assessment: design and pre-test analysis of HERO experiment, In: *Proceedings of the 26th International Conference on Nuclear Engineering (ICONE 2018)*, London, 22–26 July 2018, (2018).
- Lorusso, P., Di Piazza, I., Martelli, D., Musolesi, A., Tarantino, M., 2021b. Design of a novel test section for the LFR development: the CIRCE-THETIS facility, 28th International Conference on Nuclear Engineering (ICONE28), August 4–6, 2021, virtual conference, Volume 2, Article number V002T07A032. DOI: 10.1115/ICONE28-65575.
- Lorusso, P., Pesetti, A., Tarantino, M., Narcisi, V., Giannetti, F., Forgione, N., Del Nevo, A., 2019. Experimental analysis of stationary and transient scenarios of ALFRED steam generator bayonet tube in CIRCE-HERO facility. *Nucl. Eng. Des.* 352, 110169 <https://doi.org/10.1016/j.nucengdes.2019.110169>.
- Magni, A., Barani, T., Del Nevo, A., Pizzocci, D., Staicu, D., Van Uffelen, P., Luzzi, L., 2020. Modelling and assessment of thermal conductivity and melting behaviour of MOX fuel for fast reactor applications, *J. Nuc. Mater.* 541 (2020), 152410, pp. 1–13. DOI 10.1016/j.jnucmat.2020.152410.
- Martelli, E., Giannetti, F., Caruso, G., Tarallo, A., Polidori, M., Barucca, L., Del Nevo, A., 2018. Study of EU DEMO WCLL breeding blanket and primary heat transfer system integration. *Fusion Eng. Des.* 2018 (136), 828–833. <https://doi.org/10.1016/j.fusengdes.2018.04.016>.
- Martelli, D., Marinari, R., Barone, G., Di Piazza, I., Tarantino, M., 2017. CFD thermo-hydraulic analysis of the CIRCE fuel bundle. *Ann. Nucl. Energy* 103, 294–305.
- Meloni, P., et al., 2008. Verification of the RELAP5 Code against the MEGAPIE Irradiation Experiment, *Proceedings of the 10th OECD NEA Information Exchange Meeting on Actinide and Fission Product Partitioning and Transmutation*, OECD/NEA, Mito, Japan (2008).
- NEA-OECD, Handbook on Lead-bismuth Eutectic Alloy and Lead Properties, Materials Compatibility, Thermal-hydraulics and Technologies. NEA. No. 7268, 2015.
- NEA-OECD, Benchmarking of Thermal-Hydraulic Loop Models for Lead-Alloy-Cooled Advanced Nuclear Energy Systems - Phase II Natural Convection, Nuclear Science NEA/NSC/R(2018)1 September 2018.
- Nitti, F.S., Meloni, P., 2010. RELAP5 Code Validation in the Framework of the LACANES OECD/NEA Benchmark for HLM Innovative Nuclear System, *Proceeding NUTHOS8, Topical Meeting on Nuclear Thermal-Hydraulics, Operation and Safety*, Shanghai, China, October 10–14, 2010.
- Oriolo, F., 2000. Modifiche del Codice RELAP5 Versione MOD3.2 per la Simulazione di Sistemi Refrigerati con Leghe di Pb o Pb-Bi; RL 031/00. Università di Pisa, Pisa, Italy.
- Pesetti, A., Del Nevo, A., Forgione, N., 2015. Experimental investigation and SIMMER-III code modelling of LBE–water interaction in LIFUS5/Mod2 facility. *Nucl. Eng. Des.* 290 (2015), 119–126. <https://doi.org/10.1016/j.nucengdes.2014.11.016>.
- Shin, Y.H., et al., 2014. Cross comparison of one dimensional thermal hydraulic codes on natural circulation analysis of nacie lo op test for lead alloy cooled advanced nuclear energy systems (LACANES), *Proceedings of the 22nd International Conference on Nuclear Engineering ICONE22*, July 7–11, 2014, Prague, Czech Republic.
- Tarantino, M., Di Piazza, I., Martelli, D., Rozzia, D., Marinari, R., Pesetti, A., Lorusso, P., 2019. Design of experimental liquid metal facilities. In: *Thermal Hydraulics Aspects of Liquid Metal Cooled Nuclear Reactors*. Elsevier, pp. 83–105.
- Tarantino, M., Angiolini, M., Bassini, S., Cataldo, S., Ciantelli, C., Cristalli, C., Del Nevo, A., Di Piazza, I., Diamanti, D., Eboli, M., Fiore, A., Grasso, G., Lodi, F., Lorusso, P., Marinari, R., Martelli, D., Papa, F., Sartorio, C., Utili, M., Venturini, A., 2021a. Overview on lead-cooled fast reactor design and related technologies development in ENEA. *Energies* 2021 (14), 5157. <https://doi.org/10.3390/en1465157>.
- Tarantino, M., Forgione, N., Del Nevo, A., Bandini, G., 2012. Post Test Analysis of ICE Tests, *Proc. of 20th Int. Conf. on Nuc. Eng. (ICONE20)*, Anaheim, CA, USA, July 30–August 3, 2012, Paper No. ICONE20-54952, Vol. 2, pp. 703–712, 10 pages.
- Tarantino, M., Lorusso, P., Del Nevo, A., Di Piazza, I., Giannetti, F., Martelli, D., 2021b. Preliminary design of a helical coil steam generator mock-up for the CIRCE facility for the development of DEMO LiPb heat exchanger. *Fusion Eng. Des.* 169, 112459.
- The RELAP5 Development Team. Code Structure, System Models and Solution Methods. In RELAP5/MOD3 Code Manual; U.S. Nuclear Regulatory Commission: Washington, DC, USA, 1995.
- The RELAP5-3D Code Development Team, 2015b. RELAP5-3D© Input Data Requirements, INEEL-EXT-98-00834-V2AppA.
- The RELAP5-3D Code Development Team, 2015c. RELAP5-3D© Code Manual Vol. IV: Models and Correlations, INEEL-EXT-98-00834.
- The RELAP5-3D© Code Development Team, 2015a. RELAP5-3D© Code Manual Volume IV: Models and Correlations, INL/MIS-15-36723, Revision 4.3, Idaho, October 2015.
- Zanini, L., Dementjev, S., Gröschel, F., Leung, W., Milenkovic, R., Thomsen, K., Wagner, W., Wohlmuther, M., Cheng, X., Class, A., Konobeyev, A., Agostini, P., Meloni, P., David, J.-C., Letourneau, A., Leray, S., Panebianco, S., Cachon, L., Latgé, C., Roubin, P., Guertin, A., Thiolière, N., Dierckx, M., 2011. Experience from the post-test analysis of MEGAPIE. *J. Nucl. Mater.* 415 (3), 367–377.